

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 33 Eukaryotic Pathogens: Fungi, Protozoans, and Helminths

33.1 Multiple Choice Questions

1) Malaria is

- A) the most common cause of death due to infectious disease worldwide.
- B) prevented or treated with chloroquine.
- C) controlled by reducing the population of the mosquito vector.
- D) the most common cause of death due to infectious disease worldwide, although it can be controlled by reducing the population of the mosquito vector and prevented or treated with chloroquine.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.5

2) The initial infective stage of the malarial parasite is the

- A) sporozoite.
- B) merozoite.
- C) schizont.
- D) plasmodyite.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.5

3) Which approach is most effective for controlling malaria?

- A) vaccination with DNA vaccines
- B) vaccination with recombinant particle vaccines
- C) taking measures to reduce tick populations, such as cutting tall grass
- D) taking measures to reduce mosquito populations, such as draining swamps

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.5

4) Which of the following types of vaccines are currently in development for malaria?

- A) synthetic peptide vaccines
- B) recombinant particle vaccines
- C) DNA vaccines
- D) synthetic peptide, recombinant particle and DNA vaccines

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.5

5) The most serious types of fungal infections are

- A) subcutaneous infections.
- B) mycotoxic infections.
- C) systemic mycoses.
- D) superficial mycoses.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.1

6) A common mechanism for fungal diseases is

- A) endotoxin production.
- B) intracellular invasion.
- C) hypersensitivity reactions.
- D) red blood cell lysis.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.1

7) Many fungi produce _____ that can cause significant disease.

- A) cellulases
- B) mycotoxins
- C) endotoxins
- D) chitinases

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.1

8) Athlete's foot is an example of a

- A) protozoal infection.
- B) superficial fungal infection.
- C) superficial parasitic infection.
- D) hypersensitivity reaction.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.1

9) All fungal infections can broadly be termed

- A) benign.
- B) systemic.
- C) superficial.
- D) mycoses.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.1

10) The organism causing giardiasis is a

- A) gram-negative spirochete.
- B) flagellated protozoan.
- C) retrovirus.
- D) type of yeast.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.4

11) *Giardia* cysts are unusual because they are capable of withstanding

- A) chemical disinfection.
- B) filtration.
- C) clarification.
- D) chemical disinfection, filtration, and clarification.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.4

12) The microorganism *Cryptosporidium parvum* is a

- A) fungus.
- B) bacterium.
- C) protozoan.
- D) virus.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.4

13) *Cryptosporidium parvum* oocysts are highly resistant to

- A) heat.
- B) filtration.
- C) desiccation.
- D) chlorine.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.4

14) The treatment regimen for people with normal immunity recovering from cryptosporidiosis includes

- A) multiple antibiotics.
- B) antihelmenthic chemotherapy.
- C) multiple antibiotics and careful monitoring of fluid levels.
- D) oral rehydration only.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 33.4

15) An untreated *Entamoeba histolytica* infection can invade the

- A) liver.
- B) lungs.
- C) brain.
- D) liver, lungs, and brain.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.3

16) A common method used in the diagnosis of giardiasis is detection of *Giardia*

- A) cysts in stool.
- B) antigens in blood.
- C) antigens in stool.
- D) cysts or antigens in stool.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.4

17) *Naegleria fowleri* is a free-living _____ that causes infection by _____.

- A) amoeba / entering through the nose and burrowing into the brain
- B) fungus / entering the lungs when contaminated soil is disturbed
- C) flagellated protist / crossing the intestinal mucosa into the blood
- D) protist / entering the vagina in women, the prostate and seminal vesicles of men, or the urethra of both males and females

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.3

18) Which of the following is the most common cause of fungal infections in HIV/AIDS patients?

- A) *Candida albicans*
- B) *Giardia intestinalis*
- C) *Leishmania tropica*
- D) *Schistosoma mansoni*

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.2

19) What is the infective form of *Entamoeba histolytica* and *Giardia intestinalis*?

- A) cysts
- B) trophozoites
- C) merozoites
- D) sporozoites

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.4

20) *Trichomoniasis vaginalis* is a sexually transmitted disease that can also survive on moist surfaces and in urine for several hours. As a result, trichomoniasis

- A) is difficult to treat.
- B) is not a good candidate for eradication.
- C) can be spread by contact with contaminated toilet seats, sauna benches, and towels.
- D) is more common in males.

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 33.4

21) Roughly half of all adults in the United States are infected with _____ but are asymptomatic.

- A) *Toxoplasma gondii*
- B) *Trichomoniasis vaginalis*
- C) *Cryptosporidium parvum*
- D) *Cyclospora cayetanensis*

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.4

22) Which of the following is the most common mode of transmission of infections caused by non-fungal eukaryotic parasites?

- A) direct person-to-person contact
- B) contaminated fomites
- C) contaminated food or water
- D) contaminated needles and syringes

Answer: C

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 33.7

23) *Trypanosoma* and *Leshmania* species are transmitted by

- A) direct person-to-person contact.
- B) insect vectors.
- C) contaminated food or water.
- D) contaminated needles and syringes.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.6

24) How are trypanosome diseases similar to malaria?

- A) They are transmitted by insect vectors.
- B) They cause intermittent fevers.
- C) They are acute, self-limiting diseases.
- D) They are all endemic to Africa.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 33.6

- 25) Systemic fungal infections are most commonly transmitted by
- A) accidental inhalation of spores.
 - B) ingestion of contaminated food.
 - C) direct contact with skin lesions.
 - D) sexual contact.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.2

- 26) Which of the following is considered an opportunistic pathogen?

- A) *Toxoplasma gondii*
- B) *Trichomoniasis vaginalis*
- C) *Candida albicans*
- D) *Plasmodium vivax*

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.2

- 27) *Giardia intestinalis* and *Trichomoniasis vaginalis* lack _____ and have _____.

- A) flagella / cilia instead
- B) mitochondria / an anaerobic metabolism
- C) cell walls / rigid cell membranes instead
- D) cysts / sporozoites instead

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.4

- 28) Humans are incidental hosts for _____ and infected humans do not spread the disease.

- A) *Toxoplasma gondii*
- B) *Trichomoniasis vaginalis*
- C) *Entamoeba histolytica*
- D) *Plasmodium vivax*

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.4

- 29) Hemoflagellates that cause tropical diseases such as sleeping sickness and Chagas' disease are all transmitted by

- A) contact with fecal contaminated soil.
- B) contact with contaminated water.
- C) insect vectors.
- D) sexual contact.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.6

30) *Schistosoma* and *Plasmodium* species both require

- A) direct sexual contact for transmission.
- B) immunocompromised hosts.
- C) transmission by insect vectors.
- D) two hosts to complete their life cycle and reproduce.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.7

31) Based on the life cycle of *Schistosoma* species, which of the following would be most effective in controlling schistosomiasis?

- A) elimination of stagnate water
- B) bed nets
- C) proper sewage and water treatment
- D) frequent sanitization of shared toilets, showers, and saunas

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 33.7

32) Which of the following filariases can be controlled by thoroughly cooking pork and wild game?

- A) elephantiasis
- B) river blindness
- C) trichinosis
- D) schistosomiasis

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.7

33) Which of the following filariases are transmitted by insect vectors?

- A) elephantiasis
- B) river blindness
- C) trichinosis
- D) elephantiasis and river blindness

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.7

34) Chagas disease is caused by which pathogen?

- A) *Plasmodium vivax*
- B) *Toxoplasma gondii*
- C) *Trypanosoma cruzi*
- D) *Cryptosporidium parvum*

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.7

35) A patient arrives at the emergency room feeling extremely sick, with cycles of high fevers and chills and a severe headache. However, these symptoms are nonspecific and malaria may not be suspected if the patient is in a region where malaria is uncommon. If the medical professionals did not immediately consider malaria, which of the following test results would most strongly suggest the need for malaria testing (such as microscopic analysis of a blood smear)?

- A) the patient has a high platelet count
- B) the patient develops vomiting as well
- C) a complete blood count (CBC) shows progressively worsening severe anemia
- D) an x-ray suggests an enlarged liver and unusually small spleen

Answer: C

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 33.5

33.2 True/False Questions

1) A malarial parasite growing within erythrocytes is more resistant to the drug chloroquine than a free parasite.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 33.5

2) As a malarial infection by *Plasmodium falciparum* progresses, gametocytes are formed. The gametocytes unite in the bloodstream before being picked up by the insect vector.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.5

3) The most effective malarial control measure is to interrupt the life cycle of the parasite by eliminating the obligate host, the *Anopheles* mosquito.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.5

4) *Plasmodium vivax* causes a more serious form of malaria than does *P. falciparum*.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.5

5) Most fungi are harmless to humans, although certain superficial infections are quite common.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.1

6) *Giardia* cysts are found in most surface water sources in the United States.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.4

7) *Giardia* is especially difficult to kill because they have cell membranes that are resistant to common disinfecting chemicals.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.4

8) Eukaryotic human parasites are all respiratory aerobes.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 33.3

9) Parasitic helminths only disease in tropical regions; they do not cause disease in subtropical or temperate regions.

Answer: FALSE

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 33.7

10) Systemic fungal infections are difficult to treat because active antifungal drugs may have significant host toxicity.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.1

11) Patients with malaria often have splenomegaly.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.5

12) Anopheles mosquitoes ingest malarial gametocytes, which do not mature into gametes until transferred to the blood of a mammalian host.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.5

13) *Toxoplasma gondii* usually causes only mild illness, but can cause severe symptoms such as brain damage in some cases (especially when the host is immunocompromised).

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.4

14) The most common form of leishmaniasis is visceral leishmaniasis.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.6

15) Meningoencephalitis caused by *Naegleria fowleri* is always fatal.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.3

16) Elderly individuals are more likely to develop systemic mycoses because cell-mediated immunity generally decreases with age.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.1

17) Sporotrichosis is a subcutaneous mycosis.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.1

18) The most common major systemic mycosis in the United States is coccidiomycosis.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.2

19) *Balantidium coli* is usually transmitted by a vector.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.3

20) Trichomoniasis can be transmitted through contact with fomites, such as towels, because the causative agent can survive for a relatively long time outside of the human body.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 33.4

33.3 Essay Questions

1) How does insect ecology and the life cycle of *Plasmodium* impact the epidemiology and pathogenesis of classic malaria?

Answer: Malaria is spread only by female *Anopheles* mosquitoes, which are restricted to tropical and subtropical climates. Warm-blooded vertebrates such as humans are essential hosts for the parasite. These two factors restrict malaria to tropical and subtropical climates where significant human populations live next to or create standing water for the *Anopheles* mosquito to survive and multiply in. Fertilization occurs in the mosquito, but gamete production occurs in the human host; thus, both the mosquito and the human are required for the continued reproduction of the *Plasmodium* parasite. The malaria protist has a very complex life cycle, where it exists in many different forms such as sporozoites, merozoites, schizonts, gametes, and zygotes. A chill-fever pattern of the disease occurs when merozoites are released and lyse its host red blood cells. These symptoms alternate with asymptomatic periods when the parasite is multiplying in the red blood cells and liver.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 33.5

2) With respect to their life cycles, how are fungal pathogens different than bacterial pathogens?

Answer: Some bacterial pathogens have life cycles that depend on human infection to propagate and/or disperse the infectious agents; however, fungal pathogens do not depend on humans to thrive. Pathogenic fungi essentially have an unlimited reservoir of soil, so they are impossible to eradicate. Pathogenic bacteria often have reservoirs that limit their abundance.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 33.2

3) Fifteen high school students visited a local farm in central Wisconsin for a weekend where they enjoyed petting farm animals, riding horses, and campfire meals. With sanitation in mind, only one individual remembered to wash her hands with water pumped from a well after petting the animals before dinner. After two weeks of the girl passing out at random occasions, she was rushed into the emergency room where doctors determined she was severely dehydrated. Explain how it took direct observation of a fecal sample to solve this potential life-threatening disease and how this disease was treated in the otherwise healthy girl. Also mention any necessary precautions the individual should have taken after being diagnosed with the disease.

Answer: The protozoan pathogen *Cryptosporidium parvum* usually goes undetected from traditional clinical diagnostic tests, so the direct observation of *Cryptosporidium* oocysts in fecal matter was used to diagnose cryptosporidiosis. The disease generally subsides in healthy individuals within two weeks, so a doctor likely did not prescribe any medication but suggested to the patient to remain fully hydrated by drinking a lot of liquids. The girl likely was shedding oocysts for several weeks following infection, so she would need to be especially mindful not to spread the infectious agents to any others.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 33.4

4) Why do so many eukaryotic pathogens cause highly refractory and long-term chronic infections that are difficult to treat in contrast with prokaryotic pathogens?

Answer: Eukaryotic infections can be more difficult to treat than prokaryotic pathogens because humans (the hosts) are also eukaryotic, thus the pathogen and host share many of the same enzymes and structures. Drug targets that are unique to the pathogen can be difficult to find, and many therapeutic strategies are toxic to the host, causing harmful side effects.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 33.1

5) You and your friends are hiking the Appalachian trail in North Carolina. You run out of water and your friends want to fill up from a beautiful clear stream in the middle of the forest. What precautions should you take before drinking the water and why?

Answer: The most common disease that can be contracted from drinking untreated surface water in North America is giardiasis, caused by *Giardia intestinalis*. Even the most pristine water can contain *Giardia* cysts as wildlife such as beavers and muskrats carry the pathogen. Water should be filtered and disinfected with iodine or chlorine OR should be filtered and boiled. Camping stores carry filtering equipment and chemicals that are certified to remove or kill pathogens and make surface water safe for drinking.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 33.4

6) Why are systemic fungal infections seen only in a subset of the population, even though most people contact fungi and fungal spores in soil and dust all of the time? Contrast the epidemiology of systemic fungal infections with giardiasis, which infects virtually everyone that comes in contact with the pathogen.

Answer: Most of the population has sufficient immunity to common soil fungi to prevent disease, thus only immunocompromised people are susceptible to systemic fungal infections. In contrast, most of the population has NO natural immunity to *Giardia intestinalis*, so that any contact with the pathogen results in disease. This is why most fungi that cause systemic infections are called opportunistic pathogens.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 33.4

7) How are trypanosome diseases similar to malaria? How are they different than malaria?

Answer: Both trypanosome diseases and malaria are caused by protozoa and are spread by insect vectors. Both parasites multiply in the blood and in some cases affect the nervous system and other organs. There are no effective vaccines for trypanosome diseases or malaria. Chronic illness in either type of disease can be debilitating or deadly. They are different in that malaria has a complex life cycle and acute forms of malaria can be deadly. Trypanosome diseases are spread by tsetse flies and "kissing bugs," while malaria is spread only by the *Anopheles* mosquito.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 33.6

8) Use the events in the life cycle of *Schistosoma mansoni* to propose two different strategies to reduce the incidence of snail fever.

Answer: The life cycle of *Schistosoma mansoni* requires both snails and humans (or other mammals) as hosts. Schistosome eggs hatch in freshwater environments such as rivers or lakes and release miracidia that infect snails. Inside the snail, the miracidia transform in cercaria that are released into the environment. The cercaria can then infect humans or other mammals by burrowing into the skin and migrating to the lung and liver. The cercaria mature and produce eggs that are released in urine and feces. The eggs then hatch and the cycle continues. Separation of sewage from water used for washing, swimming, and drinking would disrupt the cycle and prevent the exposure of humans to the miracidia and the reinfection of snails from urine and feces. Another strategy would be to reduce the number of snails in freshwater environments that are used for washing and drinking water. Lastly, treatment of people with schistosomiasis would reduce the number of eggs shed into water supplies.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 33.7